



Database Forensics

How Data Solves Crimes in Large Organizations

**A 5 - slide guide to investigating data
at scale with SQL and audit logs**

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SIGN-MAGNITUDE REPRESENTATION

Sign-magnitude representation is a binary system where the MSB is used as a sign bit (0 = positive, 1 = negative), and the remaining bits represent the magnitude of the number, resulting in a range from $-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$, with **two representations for zero**.

What is a Sign Bit?

A sign bit is the leftmost bit (MSB) used to indicate whether a number is positive or negative.

0 → Positive (+)

1 → Negative (-)

What is Sign-Magnitude Representation?

Sign-magnitude representation is a method for storing signed numbers in binary where:

The first bit (MSB) is reserved as the sign bit.
The remaining bits store the magnitude (absolute value).

Rules

0 = positive

1 = negative

CONCEPT

Sign-magnitude representation is a system where:

1 bit is used for the sign

Remaining bits are used for the magnitude (value)

BIT STRUCTURE

For an n-bit number:

S | MMM

S = Sign bit

0 = positive

1 = negative

MMM = magnitude bits





4-bit sign-magnitude

Binary	Sign Bit	Magnitude	Decimal Value
0101	0	5	+5
1101	1	5	-5
0110	0	110	+6
1110	1	110	-6
0111	0	111	+7
1111	1	111	-7
1000	1	000	-0
0000	0	000	+0

Range of 4-bit Sign-Magnitude Numbers

Since one bit is reserved for the sign, only 3 bits remain for magnitude:

Positive range: +0 to +7

Negative range: -0 to -7

Notice something strange: 0 has two representations which is a flaw

What about a byte?





Range from $-(2^{n-1} - 1)$ to $+(2^{n-1} - 1)$

General rule

For an n-bit sign-magnitude number:

Sign bit = 1 bit

Magnitude bits = $n - 1$ bits

Range:

for a nibble it will be $8-1 = -7$ to $+7$

positive range $+0$ to $+7$

negative range -0 to -7

byte = $128-1 = -127$ to $+127$

positive range $+0$ to $+127$

negative range -0 to -127

n starts from 0 , 1 , 2, 3, 4 (total count of bit)

if it's a nibble $n = 4$

For signed magnitude representation
since we allocate one bit for sign it
becomes $(n-1)$

for a nibble it will be $8-1 = -7$ to $+7$

byte

$n = 8$

$= 128-1 = -127$ to $+127$





TERMS CHEAT SHEET

Term

Meaning

- | | |
|---|--|
| ☐ Sign Bit | Leftmost bit indicating sign |
| ☐ 0 sign bit | Positive number |
| ☐ 1 sign bit | Negative number |
| ☐ Magnitude | Absolute value stored in the remaining bits |
| ☐ Sign-Magnitude Representation
magnitude | A system that uses one bit for sign and the rest for |
-
- ☐ The first bit tells the sign, and the rest of the bits tell the magnitude.
 - ☐ No, there is no confusion in ON/OFF states. The hardware only deals with 0 (OFF) and 1 (ON).
 Signed or unsigned interpretation is applied later by the system, not by the hardware itself.
 - ☐ ON/OFF is physical reality
 - ☐ Signed/unsigned is logical interpretation
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- ☐ Unsigned = "all bits = value"
 - ☐ Sign-magnitude = "1 bit = sign, rest = value"

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